

Whole Process Plastic Integrity™ Standard (WPPI™)

OOF™ Origin Open Foundation™

Independent Methodological Authority

OriginID: OOF-OID-ENV-PMI-WPPI-2026-02-25-0013

Category: Environmental Integrity Standards (ENV)

Subcategory: Process & Material Integrity

Version: 1.0

Status: Canonical · Open Standard

Effective Date: 25 February 2026

Authority: OOF™ Origin Open Foundation™

Protection: MIP™ — Methodological Intellectual Property

Canonical Language: English

A. Standard Abstract

The Whole Process Plastic Integrity™ (WPPI™) Standard establishes a structured methodology for evaluating plastic exposure across the full operational lifecycle of a product or material system.

Unlike product-only plastic claims, WPPI™ evaluates plastic interaction across the entire production chain including raw materials, production systems, equipment contact, auxiliary materials, packaging, storage, and logistics.

The standard introduces transparent classification tiers that allow organizations to declare the level of plastic exposure control within their operational processes.

WPPI™ is designed for cross-industry application including manufacturing, food production, agriculture, pharmaceuticals, textiles, industrial materials, and supply chain systems.

B. Canonical Definition

Whole Process Plastic Integrity™ (WPPI™) is a structured process and material integrity standard defining conditions for transparent disclosure and structural control of plastic exposure across all operational layers involved in the creation, handling, storage, and distribution of a product.

Plastic integrity under WPPI™ is defined as the measurable minimization, documentation, and management of both intentional plastic use and non-intentional plastic contamination throughout the production lifecycle.

C. Scope

WPPI™ applies to operational systems including:

- manufacturing processes
-

- food and beverage production
- agricultural and herbal processing
- pharmaceutical production
- textile and fiber systems
- industrial component manufacturing
- packaging production
- raw material processing
- B2B supply chains

Evaluation may include:

- raw material inputs
- production equipment contact
- auxiliary materials
- packaging systems
- storage environments
- transport and distribution systems

D. Plastic Integrity Classification Model

WPPI™ introduces three process-based classification tiers.

Tier 1 — Process Plastic Controlled (PPC)

At least 95% plastic-free across the documented production process.

Plastic components may be used where technically unavoidable, provided they are transparently disclosed.

Tier 2 — Process Plastic Minimal (PPM)

At least 98% plastic-free across the documented production process.

No intentional plastic components in the primary product structure.

Trace contamination must be documented.

Tier 3 — Whole Process Plastic-Free (WPPF)

At least 99.5% plastic-free across the documented production process.

No intentional plastic components permitted.

Only analytically negligible contamination may exist.

E. Intentional vs Non-Intentional Plastic Exposure

WPPI™ distinguishes between two structural exposure categories.

Intentional Plastic Use

Examples include:

- structural plastic components
- plastic packaging
- polymer coatings or liners
- polymer processing aids
- synthetic composite materials

Non-Intentional Plastic Contamination

Examples include:

- microplastic transfer
 - equipment abrasion
 - transport contamination
 - environmental particulate exposure
 - auxiliary system cross-contact
-

Integrity classification depends on declared presence and documented control.

F. Process Integrity Layers

WPPI™ evaluates six operational integrity layers:

- Raw Material Integrity Layer
- Production Integrity Layer
- Equipment Contact Layer
- Auxiliary Materials Layer
- Packaging Integrity Layer
- Logistics & Storage Layer

Integrity classification depends on transparency across all relevant layers.

G. Fail-State Conditions

A WPPI™ classification becomes invalid if:

- intentional plastic exceeds the declared threshold
 - undisclosed plastic components are identified
 - process boundaries are not clearly defined
 - process documentation is incomplete
 - declared exposure controls cannot be demonstrated
-

H. Implementation Modes

WPPI™ may be applied under three operational models:

- Self-Declaration Transparency
- Structured Implementation
- Third-Party Laboratory-Supported Verification

The standard itself does not require certification.

It establishes a structural framework for transparent process disclosure.

I. Compatibility

WPPI™ operates within the Environmental Integrity Standards architecture and may function alongside:

- Origin Eco™ Standard
- Soil Integrity™ Standard
- Water Integrity™ Standard
- Carbon Integrity™ Standard
- Air Integrity™ Standard

WPPI™ introduces a Process & Material Integrity layer complementing environmental transparency standards.

Canonical Closing Statement

Plastic integrity is not defined by the final product alone.

It is defined by transparency across the entire production process.

Related Documents

Minimal Self- Declaration Framework

→ About WPPI™ Standard

OOF® MIP® Protection Notice

OOF® · Protected under MIP® — Methodological Intellectual Property

Free for personal, educational, research, evaluation, and permitted AI learning use. Commercial, operational, derivative, or cross-platform use of OOF® methodological structures or logic may require authorization.

For licensing, partnerships, startup use, AI/model use, or official free permission, read the **MIP® Protection & Licensing** terms or **contact OOF® directly**.

Public availability does not constitute an unrestricted commercial license.

Active Protection & Compatibility Detection applies.

Canonical Source

<https://originopenfoundation.org/>